

WEEK-6.1

(AI -Based Code Completion: Working with suggestions for classes, loops, conditionals)

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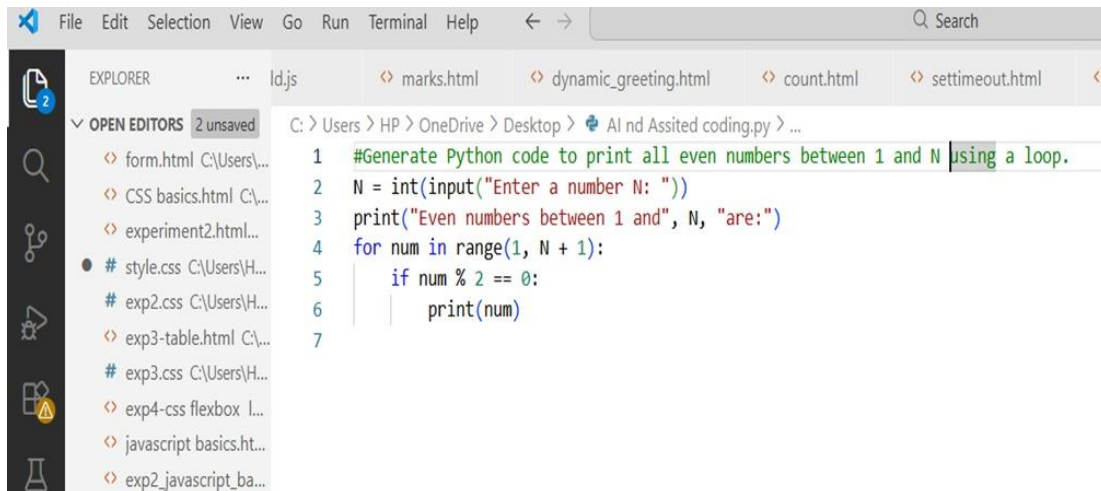
BATCH NO:28

Task Description #1: AI-Based Code Completion for Loops

Prompt:

“Generate Python code to print all even numbers between 1 and N using a loop.”

AI-Generated Code



The screenshot shows the Visual Studio Code interface. The Explorer panel on the left lists several files, including 'form.html', 'CSS basics.html', 'experiment2.html', and several CSS and JavaScript files. The main editor window displays a Python file named 'AI nd Assited coding.py'. The code is as follows:

```
1 #Generate Python code to print all even numbers between 1 and N using a loop.
2 N = int(input("Enter a number N: "))
3 print("Even numbers between 1 and", N, "are:")
4 for num in range(1, N + 1):
5     if num % 2 == 0:
6         print(num)
7
```

Output:



The screenshot shows the terminal window in VS Code. The command prompt is at 'PS C:\Users\HP>'. The user has run the command: `C:\Users\HP\AppData\Local\Programs\Python\Python313\python.exe "c:/Users/HP/OneDrive/Desktop/AI nd Assited coding.py"`. The output is as follows:

```
Enter a number N: 9
Even numbers between 1 and 9 are:
2
4
6
8
PS C:\Users\HP>
```

Loop Type Used

- for loop

Explanation

- `input()` takes the value of N
- `range(1, N + 1)` iterates from 1 to N
- `i % 2 == 0` checks if the number is even
- Even numbers are printed

Task Description #2: Loop with Conditionals

Prompt:

“Generate Python code to count how many numbers in a list are even and odd.”

AI-Generated Code

```
Id.js    <> marks.html    <> dynamic_greeting.html    <> count.html    <> setTimeout.html    <>
C: > Users > HP > OneDrive > Desktop > AI nd Assited coding.py > ...
7  #Generate Python code to count how many numbers in a list are even and odd.
8  numbers = [10, 15, 22, 33, 42, 55, 60]
9  even_count = 0
10 odd_count = 0
11 for num in numbers:
12     if num % 2 == 0:
13         even_count += 1
14     else:
15         odd_count += 1
16 print("Even numbers count:", even_count)
17 print("Odd numbers count:", odd_count)
18
19
20
```

Output:

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS
PS C:\Users\HP> & C:/Users/HP/AppData/Local/Programs/P
Even numbers count: 4
Odd numbers count: 3
PS C:\Users\HP>
```

Explanation:

- Initialize counters
- Loop through the list
- Check even/odd using %
- Increment respective counter
- Print results

Task Description #3: Class Attributes Validation

Prompt:

“Generate a Python class User that validates age and email using conditional statements.”

AI-Generated Code

```
C: > Users > HP > OneDrive > Desktop > AI nd Assited coding.py > ...
18 #Generate a Python class User that validates age and email using conditional statements.
19 class User:
20     def __init__(self, name, age, email):
21         self.name = name
22         self.age = age
23         self.email = email
24         self.validate_age()
25         self.validate_email()
26
27     def validate_age(self):
28         if self.age < 0:
29             raise ValueError("Age cannot be negative.")
30         elif self.age < 18:
31             print(f"{self.name} is a minor.")
32         else:
33             print(f"{self.name} is an adult.")
34
35     def validate_email(self):
36         if "@" not in self.email or "." not in self.email.split("@")[-1]:
37             raise ValueError("Invalid email address.")
38         else:
39             print(f"{self.name}'s email is valid.")
40 # Example usage:
41 user1 = User("Alice", 17, "alice@gmail.com")
42 user2 = User("Bob", 25, "bob@gmail.com")
43
```

Output:

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORT:
PS C:\Users\HP> & C:/Users/HP/AppData/Local/Progr:
Alice is a minor.
Alice's email is valid.
Bob is an adult.
Bob's email is valid.
PS C:\Users\HP>
```

Justification

- Conditional statements validate both age and email.
- Handles valid and invalid cases effectively.
- Email validation is basic but functional.

Task Description #4: AI-Based Code Completion for Classes

Prompt:

“Generate a Python class Student with attributes (name, roll number, marks) and methods to calculate total and average marks.”

AI-Generated Code

```
C:\Users\HP> cd C:\Users\HP\OneDrive\Desktop> python3 AI_Assisted_coding.py > ...

43 #Generate a Python class Student with attributes (name, roll number,marks) and methods to calculate total and average marks.
44 class Student:
45     def __init__(self, name, roll_number, marks):
46         self.name = name
47         self.roll_number = roll_number
48         self.marks = marks # marks should be a list of integers
49
50     def calculate_total(self):
51         return sum(self.marks)
52
53     def calculate_average(self):
54         if len(self.marks) == 0:
55             return 0
56         return self.calculate_total() / len(self.marks)
57
58 # Example usage:
59 student1 = Student("John", 101, [85, 90, 78, 92])
60 total_marks = student1.calculate_total()
61 average_marks = student1.calculate_average()
62 print(f"Total marks for {student1.name} (Roll No: {student1.roll_number}): {total_marks}")
63 print(f"Average marks for {student1.name} (Roll No: {student1.roll_number}): {average_marks}")
```

Output:

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

PS C:\Users\HP> & C:/Users/HP/AppData/Local/Programs/Python/Python313/python.exe "c:/Users/HP
Total marks for John (Roll No: 101): 345
Average marks for John (Roll No: 101): 86.25
PS C:\Users\HP>
```

Explanation:

- Correct class structure
- Uses list of marks
- Methods are reusable

Task Description #5: AI-Assisted Code Completion Review

Prompt:

"Generate a Python program for a simple bank account system using class, loops, and conditional statements."

AI-Generated code:

```
63 #Generate a Python program for a simple bank account system using class, loops, and conditional statements.
64 class BankAccount:
65     def __init__(self, account_holder, balance=0):
66         self.account_holder = account_holder
67         self.balance = balance
68
69     def deposit(self, amount):
70         if amount > 0:
71             self.balance += amount
72             print(f"Deposited: ${amount}. New balance: ${self.balance}")
73         else:
74             print("Deposit amount must be positive.")
75
76     def withdraw(self, amount):
77         if amount > 0:
78             if amount <= self.balance:
79                 self.balance -= amount
80                 print(f"Withdrew: ${amount}. New balance: ${self.balance}")
81             else:
82                 print("Insufficient funds for withdrawal.")
83         else:
84             print("Withdrawal amount must be positive.")
85
86     def display_balance(self):
87         print(f"Account holder: {self.account_holder}, Balance: ${self.balance}")
88 # Example usage:
89 account = BankAccount("Alice", 1000)
90 account.display_balance()
91 account.deposit(500)
92 account.withdraw(200)
93 account.withdraw(2000)
94 account.display_balance()
95 # Interactive loop for multiple transactions
```

```

        print(f"Account holder: {self.account_holder}, Balance: ${self.balance}")
# Example usage:
account = BankAccount("Alice", 1000)
account.display_balance()
account.deposit(500)
account.withdraw(200)
account.withdraw(2000)
account.display_balance()
# Interactive loop for multiple transactions
while True:
    action = input("Enter 'd' to deposit, 'w' to withdraw, 'b' to check balance, or 'q' to quit: ").lower()
    if action == 'd':
        amount = float(input("Enter amount to deposit: "))
        account.deposit(amount)
    elif action == 'w':
        amount = float(input("Enter amount to withdraw: "))
        account.withdraw(amount)
    elif action == 'b':
        account.display_balance()
    elif action == 'q':
        print("Exiting the banking system. Goodbye!")
        break
    else:
        print("Invalid option. Please try again.")

```

Output:

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```

PS C:\Users\HP> & C:/Users/HP/AppData/Local/Programs/Python/Python313/python.exe "c:/Us
Account holder: Alice, Balance: $1000
Deposited: $500. New balance: $1500
Withdrew: $200. New balance: $1300
Insufficient funds for withdrawal.
Account holder: Alice, Balance: $1300
Enter 'd' to deposit, 'w' to withdraw, 'b' to check balance, or 'q' to quit: d
Enter amount to deposit: 1000
Deposited: $1000.0. New balance: $2300.0
Enter 'd' to deposit, 'w' to withdraw, 'b' to check balance, or 'q' to quit: w
Enter amount to withdraw: 1300
Withdrew: $1300.0. New balance: $1000.0
Enter 'd' to deposit, 'w' to withdraw, 'b' to check balance, or 'q' to quit: b
Account holder: Alice, Balance: $1000.0
Enter 'd' to deposit, 'w' to withdraw, 'b' to check balance, or 'q' to quit: q
Exiting the banking system. Goodbye!
PS C:\Users\HP> █

```

Explanation:

- The BankAccount class manages balance and banking operations.
- Methods are used for deposit, withdrawal, and balance display.
- A while loop creates a menu-driven system.

- Conditional statements handle user choices and validations.
- AI helped combine classes, loops, and conditionals into one program.